Mathematical and Computational Methods

Exercise Sheet 2: Complex numbers I: algebra and geometry

This sheet accompanies *Unit 2 (Complex numbers I: algebra and geometry)*. Every skill of the unit is exercised: Cartesian arithmetic, the conjugate and modulus, quadratics with complex roots, the Argand diagram, and the polar form. Work in radians throughout.

How to use this sheet. Problems are graded by difficulty—$\star$ drill (routine fluency), $\star\star$ core (standard, tutorial level) and $\star\star\star$ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit2}"`) and recompile.

Reminders. For $z = a + b\,\mathrm{i}$: conjugate $\bar{z} = a - b\,\mathrm{i}$, modulus $|z| = \sqrt{a^2 + b^2}$ with $|z|^2 = z\bar{z} = a^2 + b^2$, and division by $w \neq 0$ via $z/w = z\,\overline{w}/\,|w|^2$. Polar form $z = r(\cos\theta + \mathrm{i}\sin\theta)$ with $r = |z|$ and $\theta = \arg z$ (principal value $\operatorname{Arg} z \in (-\pi, \pi]$, fixed by the quadrant); products multiply moduli and add arguments.

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1 Warm-up drills

Exercise 1.1 *($\star$ drill— powers of* i*)*. Simplify the powers of the imaginary unit

$$\mathrm{i}^3, \qquad \mathrm{i}^4, \qquad \mathrm{i}^{15}, \qquad \mathrm{i}^{2026}, \qquad \mathrm{i}^{-1}, \qquad \mathrm{i}^{-7}.$$

Exercise 1.2 *($\star$ drill— Cartesian arithmetic)*. With $z = 3 + 2\mathrm{i}$ and $w = 1 - 5\mathrm{i}$, find $z + w$, $z - w$, zw and z^2.

Exercise 1.3 *($\star$ drill— conjugate and modulus)*. For $z = 12 - 5\mathrm{i}$ write down $\bar{z}$ and compute $|z|$ and $z\,\bar{z}$.

Exercise 1.4 *($\star$ drill— real and imaginary parts)*. For $z = -4 + 7\mathrm{i}$ state $\operatorname{Re} z$, $\operatorname{Im} z$, $\operatorname{Re}(\bar{z})$ and $\operatorname{Im}(\mathrm{i}z)$.

Exercise 1.5 *($\star$ drill— division)*. Write $\dfrac{4 + 3\mathrm{i}}{1 + 2\mathrm{i}}$ in Cartesian form $a + b\,\mathrm{i}$.

Exercise 1.6 *($\star$ drill— quadratics over $\mathbb{C}$)*. Solve $x^2 + 1 = 0$ and $x^2 + 25 = 0$ over $\mathbb{C}$.

Exercise 1.7 *($\star$ drill— to polar form)*. Convert to polar form $r(\cos\theta + \mathrm{i}\sin\theta)$ with principal argument: $1 + \mathrm{i}$, $\quad 2\mathrm{i}$, $\quad -3$.

Exercise 1.8 *($\star$ drill— back to Cartesian)*. Convert $z = 2\big(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\big)$ to Cartesian form.

Exercise 1.9 *($\star$ drill— the polar rules)*. Let $z = 3\big(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\big)$ and $w = 2\big(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\big)$. Using the polar rules, find zw and z/w in Cartesian form.

Exercise 1.10 *($\star$ drill— an Argand sketch)*. On a single Argand diagram, plot the points $z_1 = 3 + \mathrm{i}$, $z_2 = -2 + 2\mathrm{i}$ and $z_3 = -1 - 3\mathrm{i}$, together with the sum $z_1 + z_2$. Label each point.

2 Tutorial problems

Problems marked ▶ are the core set for the 2-hour tutorial; the rest are for completion at home.

Exercise 2.1 *(▶ ⋆⋆ core— the conjugate and modulus rules)*. Let $z = 2 + 3\mathrm{i}$ and $w = 1 - 4\mathrm{i}$ (as in the notes).

(a) Compute zw and z/w in Cartesian form.

(b) Verify the conjugate rule $\overline{zw} = \overline{z}\,\overline{w}$.

(c) Verify the modulus rule $|zw| = |z|\,|w|$.

Exercise 2.2 *(⋆⋆ core— a general proof (maths))*. *(Maths stream — a general proof.)* Let $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i}$ with $a, b, c, d \in \mathbb{R}$. Prove the conjugate-of-a-product rule $\overline{zw} = \overline{z}\,\overline{w}$ directly from the definition $\overline{a + b\mathrm{i}} = a - b\mathrm{i}$.

Exercise 2.3 *(▶ ⋆⋆ core— conjugate pairs)*. Solve over $\mathbb{C}$, and confirm that the roots form a conjugate pair:

(a) $x^2 + 2x + 10 = 0$; (b) $x^2 - 4x + 13 = 0$; (c) $2x^2 + x + 1 = 0$.

Exercise 2.4 *(⋆⋆ core— solve for real x, y)*. Find the real numbers x, y satisfying $(x + y\,\mathrm{i})(3 - \mathrm{i}) = 7 + \mathrm{i}$.

Exercise 2.5 *(▶ ⋆⋆ core— plotting and distance)*. Let $z = 3 + 2\mathrm{i}$ and $w = -1 + 2\mathrm{i}$. Plot $z,\ w,\ \overline{z},\ z + w,\ z - w$ on an Argand diagram, and compute the distance $|z - w|$ between the points z and w.

Exercise 2.6 *(▶ ⋆⋆ core— AC impedance (physics))*. *(Physics — AC circuits.)* A series circuit has complex impedance $Z = 4 + 3\mathrm{i}$ (in ohms). It is driven by a voltage $V = 10$ (volts, taken as a real reference).

(a) Find $|Z|$ and the phase angle $\operatorname{Arg} Z$.

(b) Compute the current $I = V/Z$ in Cartesian form and its magnitude $|I|$.

Exercise 2.7 *(⋆⋆ core— distance between records (data science))*. *(Data science — distance between records.)* Two data points in the plane are encoded as complex numbers $p = 2 + 5\mathrm{i}$ and $q = 6 + 2\mathrm{i}$. Compute the Euclidean distance $|p - q|$ between them and the midpoint $\frac{1}{2}(p + q)$.

Exercise 2.8 *(▶ ⋆⋆ core— principal argument)*. Convert each number to polar form, giving the *principal* argument $\operatorname{Arg} z \in (-\pi, \pi]$. Sketch each to fix the quadrant.

$$\text{(a)}\ -1 + \mathrm{i}; \qquad \text{(b)}\ -2 - 2\sqrt{3}\,\mathrm{i}; \qquad \text{(c)}\ \sqrt{3} - \mathrm{i}; \qquad \text{(d)}\ -5.$$

Exercise 2.9 *(⋆⋆ core— polar product and quotient)*. Let $z = \sqrt{2}\big(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\big)$ and $w = 2\big(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\big)$. Find zw and z/w, giving each in both polar and Cartesian form.

Exercise 2.10 *(▶ ⋆⋆ core— multiplication by i)*. (a) For $z = 2 + \mathrm{i}$, compute $\mathrm{i}z$ and confirm geometrically that multiplying by i is a quarter-turn anticlockwise that preserves length.

(b) Which single complex number rotates a point by $\frac{\pi}{2}$ anticlockwise *and* doubles its distance from the origin?

(c) Apply a clockwise quarter-turn to $3 + \mathrm{i}$ (i.e. multiply by the number of modulus 1 and argument $-\frac{\pi}{2}$).

Exercise 2.11 *($\star\star$ core— powers of $1+\mathrm{i}$)*. Compute $(1+\mathrm{i})^2$ and $(1+\mathrm{i})^4$ by direct multiplication, and hence find $\left|(1+\mathrm{i})^{10}\right|$ without expanding.

Exercise 2.12 *(▶ $\star\star$ core— true or false)*. *(Conceptual — true or false, with a one-line reason.)*

(a) For every $z \in \mathbb{C}$, $z^2 \geq 0$.

(b) $|z| = |\bar{z}|$ for all z.

(c) If $z + \bar{z} = 0$ then z is purely imaginary or zero.

(d) $\operatorname{Re}(zw) = \operatorname{Re}(z)\operatorname{Re}(w)$ for all z, w.

(e) There is an ordering $<$ on $\mathbb{C}$ compatible with its arithmetic — that is, one respecting $+$ and $\times$ (if $x > 0$ and $y > 0$ then $x + y > 0$ and $xy > 0$), so that every nonzero square is positive.

Exercise 2.13 *($\star\star\star$ challenge— spot the error)*. *(Spot the error.)* A student writes the following two lines. Find and correct each mistake.

$$\text{(i)}\quad \frac{2+3\mathrm{i}}{1+\mathrm{i}} = \frac{2}{1} + \frac{3}{1}\mathrm{i} = 2 + 3\mathrm{i}; \qquad \text{(ii)}\quad \sqrt{-4}\,\sqrt{-9} = \sqrt{(-4)(-9)} = \sqrt{36} = 6.$$

3 Further practice (home)

Exercise 3.1 *($\star\star$ core— a sum of powers of i)*. Evaluate the geometric-looking sum $\sum_{k=0}^{2026} \mathrm{i}^k = 1 + \mathrm{i} + \mathrm{i}^2 + \cdots + \mathrm{i}^{2026}$.

Exercise 3.2 *($\star$ drill— Cartesian arithmetic)*. Compute in Cartesian form: (a) $(3-2\mathrm{i}) + (-1+5\mathrm{i})$; (b) $(2+\mathrm{i})(2-\mathrm{i})$; (c) $(1-\mathrm{i})^2$; (d) $\mathrm{i}(3-\mathrm{i})$.

Exercise 3.3 *($\star$ drill— conjugate, modulus, reciprocal)*. For $z = -3 + 4\mathrm{i}$ find $\bar{z}$, $|z|$, $\dfrac{1}{z}$ and $\dfrac{z}{\bar{z}}$.

Exercise 3.4 *($\star\star$ core— division)*. Simplify to Cartesian form: (a) $\dfrac{1+2\mathrm{i}}{3-4\mathrm{i}}$; (b) $\dfrac{\mathrm{i}}{1+\mathrm{i}}$.

Exercise 3.5 *($\star\star$ core— quadratics over $\mathbb{C}$)*. Solve over $\mathbb{C}$: (a) $x^2 + 9 = 0$; (b) $x^2 - 6x + 25 = 0$; (c) $4x^2 + 9 = 0$; (d) $x^2 + x + 1 = 0$.

Exercise 3.6 *($\star\star$ core— a real cubic from one root)*. Given that $1 + 2\mathrm{i}$ is a root of $x^3 - 4x^2 + 9x - 10 = 0$ (real coefficients), find the other two roots.

Exercise 3.7 *($\star\star$ core— modulus of a quotient)*. Without converting to Cartesian form, evaluate $\left|\dfrac{(3+4\mathrm{i})(5-12\mathrm{i})}{1+\mathrm{i}}\right|$.

Exercise 3.8 *($\star\star$ core— loci in the plane)*. Describe geometrically (and sketch) the set of points $z \in \mathbb{C}$ satisfying

$$\text{(a)}\ |z| = 2; \qquad \text{(b)}\ |z-1| = 3; \qquad \text{(c)}\ |z-\mathrm{i}| = |z+\mathrm{i}|.$$

Exercise 3.9 *($\star\star$ core— regions in the plane)*. Sketch each *region* in the Argand plane (shade it, and show clearly whether the boundary is included):

$$\text{(a)}\ \{\, z : |z-1| < 2 \,\}; \qquad \text{(b)}\ \{\, z : 1 \leq |z| \leq 2 \,\}; \qquad \text{(c)}\ \{\, z : \operatorname{Re} z \geq 0,\ \operatorname{Im} z \geq 0 \,\}.$$

Exercise 3.10 *($\star\star$ core— polar to Cartesian)*. Convert to Cartesian form:

$$\text{(a) } 6\big(\cos\tfrac{\pi}{6} + \mathrm{i}\sin\tfrac{\pi}{6}\big); \qquad \text{(b) } 2\big(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3}\big);$$
$$\text{(c) } 5(\cos\pi + \mathrm{i}\sin\pi); \qquad \text{(d) } \sqrt{2}\big(\cos(-\tfrac{\pi}{4}) + \mathrm{i}\sin(-\tfrac{\pi}{4})\big).$$

Exercise 3.11 *($\star\star$ core— polar product and quotient)*. Let $z = 4\big(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\big)$ and $w = 2\big(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\big)$. Find zw and z/w in Cartesian form.

Exercise 3.12 *($\star\star$ core— one product, two ways)*. Compute the product $(1+\mathrm{i})(\sqrt{3}+\mathrm{i})$ in two ways — by Cartesian expansion and by the polar rule — and hence write down its modulus and argument.

Exercise 3.13 *($\star\star$ core— a dot product from complex numbers (data science))*. *(Data science.)* Encode two real 2-vectors as $u = 3 + 4\mathrm{i}$ and $v = 4 + 3\mathrm{i}$. Show that the dot product of $(3,4)$ and $(4,3)$ equals $\mathrm{Re}(u\,\overline{v})$, and find $|u\,\overline{v}|$.

Exercise 3.14 *($\star\star$ core— phasors (physics))*. *(Physics — phasors / rotating vectors.)* Two voltage phasors of the same frequency are $V_1 = 6 + 8\mathrm{i}$ and a second phasor V_2 of equal amplitude but rotated a quarter-turn (90°) anticlockwise, i.e. $V_2 = \mathrm{i}V_1$. Find V_2 and the resultant $V = V_1 + V_2$ in Cartesian form, the resultant amplitude $|V|$, and show that $|V| = \sqrt{2}\,|V_1|$.

Exercise 3.15 *($\star\star\star$ challenge— the parallelogram law)*. Prove the *parallelogram law*: for all $z, w \in \mathbb{C}$,

$$|z+w|^2 + |z-w|^2 = 2\,|z|^2 + 2\,|w|^2 .$$

Exercise 3.16 *($\star\star\star$ challenge— a square root in Cartesian form)*. Find all $z = x + y\,\mathrm{i}$ (with $x, y \in \mathbb{R}$) such that $z^2 = 3 + 4\mathrm{i}$, working directly in Cartesian form.

Exercise 3.17 *($\star\star$ core— solving with $\overline{z}$)*. Find all complex z satisfying $z + 2\overline{z} = 6 - 3\mathrm{i}$.

Exercise 3.18 *($\star\star\star$ challenge— an equilateral triangle)*. *(Rotation/geometry.)* Two vertices of an equilateral triangle lie at 0 and 2 on the real axis. Use multiplication by a complex number of modulus 1 to find the two possible positions of the third vertex.

Exercise 3.19 *($\star\star$ core— spot the error)*. *(Spot the error.)* A student claims "$|3+4\mathrm{i}| = 3 + 4 = 7$" and "$\mathrm{Arg}(-1-\mathrm{i}) = \arctan(1) = \frac{\pi}{4}$". Correct both.

Exercise 3.20 *($\star\star$ core— modulus and argument round-up)*. Suppose $|z| = 2$, $\mathrm{Arg}\,z = \frac{\pi}{3}$, $|w| = 3$, $\mathrm{Arg}\,w = \frac{\pi}{4}$. Find the modulus and argument of zw, z/w, z^2 and $\overline{z}$; convert z and z^2 to Cartesian form.